# Security Architecture

Unified Funeral Services Management Platform

## Perimeter Security

- **WAF** — Web App Firewall
- **DDoS Protection**
- **API Gateway**
- **Bot Detection**
- **SSL/TLS** — Termination

## Identity & Access Management (FSD-12)

- **SSO** — SAML / OIDC
- **MFA** — TOTP / SMS
- **RBAC** — Role-Based
- **ABAC** — Attribute-Based
- **PAM** — Privileged Access

### User Directory & Federation

Internal Users (LDAP/AD) • External Users (Social / Partner IdP) • Service Accounts

## Application Security

- Input Validation
- Session Management
- CSRF / XSS Protection
- SQL Injection Prevention
- Secure Headers
- Secure Cookies
- Dependency Scanning
- Secrets Management — Vault
- API Security
- File Upload Security

## Data Security (FSD-28)

### Encryption at Rest

- AES-256
- Database TDE
- File Encryption
- Backup Encryption

### Encryption in Transit

- TLS 1.3
- mTLS
- VPN

### Key Management

- HSM
- Key Rotation
- Key Custody
- Key Escrow

### Data Masking

- PII Masking
- Dynamic Masking
- Static Masking

### Tokenization

- Payment Tokens
- PII Tokenization
- Format Preserving

### Privacy Controls

- GDPR / CCPA
- Consent Mgmt
- Right to Erasure

## Audit & Compliance (FSD-27)

- **Immutable Audit Logs** — Blockchain-inspired
- **User Activity Tracking**
- **Data Access Logging**
- **Compliance Reporting** — GDPR / CCPA / HIPAA
- **Forensic Analysis** — SIEM

## Threat Detection & Response

- **SIEM** — Splunk / ELK
- **IDS / IPS** — Network & Host
- **Anomaly Detection** — UEBA
- **Incident Response** — SOAR
- **Vulnerability Management** — Scanning